

MOHITH PENUMURU

AI & Data Engineer — Python, Databricks, PySpark, AWS, LLMs, Snowflake, SQL

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PROFESSIONAL SUMMARY

AI & Data Engineer with 3+ years of experience building production-grade data pipelines and AI-powered solutions on AWS and Snowflake. Skilled in designing scalable ETL/ELT systems, developing intelligent document processing and reconciliation workflows using AWS Bedrock and LangChain, LangGraph and delivering real-time analytics across cloud environments. Proven track record of automating complex data workflows, integrating Generative AI into enterprise platforms, and transforming raw data into actionable insights that drive business decisions.

EXPERIENCE

Deloitte

Sept 2025 – Present

AI & Data Engineer — Bengaluru, India

- Designed and deployed LLM-powered data intelligence solutions using AWS Bedrock (Agent Runtime, Agent Core), LangChain, and LangGraph for automated document understanding and multi-source reconciliation workflows.
- Built intelligent invoice reconciliation pipelines that process documents across multiple formats (JPG, PNG, PDF) using AWS services and Bedrock, matching records from 10+ sources with 96% accuracy.
- Developed conversational analytics interfaces using LangChain and LangGraph, enabling non-technical stakeholders to query complex datasets through natural language prompts without writing SQL.
- Engineered end-to-end data pipelines on AWS with PySpark and Delta Lake, processing 100GB+ daily data for downstream AI model consumption.
- Orchestrated multi-step AI agent workflows with LangGraph for complex reasoning tasks including data validation, anomaly detection, and automated report generation.
- Collaborated with business teams to translate non-technical requirements into scalable AI solutions, reducing manual data reconciliation effort by 60%.

Accenture

Jan 2023 – Sept 2025

Data Engineering, Management & Governance Analyst — Bengaluru, India

- Developed and automated ETL pipelines using AWS Glue, EMR, Lambda, EventBridge, Step Functions, and Airflow to extract and transform data from S3, Data Lake, and Delta Lake.
- Utilized Databricks with PySpark for data aggregation, processing, and building machine learning models across stock, cricket, and taxi datasets.
- Built an end-to-end ELT pipeline in Snowflake using a medallion architecture (Bronze → Silver → Gold), ingesting raw data files via external stages into bronze tables, applying filtering and joins in silver, and performing aggregations/pivoting in gold — orchestrated through Python stored procedures scheduled with Snowflake Tasks (cron jobs).
- Optimized data pipelines using Spark SQL for real-time analytics and high-performance querying, reducing average runtime by 40%.
- Built interactive dashboards and managed Salesforce Marketing Cloud (SFMC) data extensions and automations. Worked with Amazon Redshift and Amazon RDS for large-scale data storage and transactional processing.
- Managed CI/CD workflows and infrastructure-as-code using GitHub, GitHub Actions, Jenkins, Groovy scripts, and Terraform.
- Conducted extensive performance tuning and refactored ETL jobs to reduce runtime and compute cost by 30%.

- Implemented automations to streamline workflows and minimize manual intervention, improving productivity and data reliability.

PROJECTS

LLM-Powered Data Intelligence & Reconciliation Platform

(AWS Bedrock, LangChain, LangGraph, AWS Strands, Terraform, GitHub Actions)

- Built an LLM-based natural language querying interface using LangChain and AWS Bedrock, enabling non-technical stakeholders to query Snowflake and Delta Lake datasets through conversational prompts.
- Designed multi-agent reconciliation workflows with LangGraph and Bedrock Agent Runtime to automatically match and validate invoices across multiple formats (JPG, PNG, PDF) from 10+ disparate sources.
- Implemented prompt engineering and retrieval-augmented generation (RAG) with Bedrock Agent Core to improve query accuracy by 35% over baseline text-to-SQL approaches.
- Reduced manual reconciliation time by 60% and ad-hoc data request volume by 45% by empowering business teams with self-serve AI-driven analytics.
- Automated CI/CD pipelines using GitHub Actions for model deployment and infrastructure updates, with AWS resources provisioned via Terraform under full version control.

Enterprise Data Engineering Platform

(Databricks, PySpark, AWS Glue, Talend, Snowflake, Airflow, Jenkins, Terraform)

- Designed scalable ETL pipelines using Talend and AWS Glue to ingest data from 12+ sources into S3 and Delta Lake with automated schema evolution.
- Built distributed transformation workflows on Databricks with PySpark, processing 200M+ records daily with data quality checks and lineage tracking.
- Deployed orchestration pipelines via Airflow and AWS Step Functions, reducing manual scheduling overhead by 70%.
- Automated CI/CD pipelines using Jenkins for Databricks notebook deployments, and provisioned AWS resources (Redshift, S3, Lambda) via Terraform, reducing provisioning time by 60%.

SKILLS

Programming	Python, SQL, PySpark, Spark SQL, Shell Scripting
AI & Generative AI	LLMs, LangChain, LangGraph, RAG, AWS Bedrock, Prompt Engineering, Text-to-SQL
Machine Learning	Model Development, Feature Engineering, Predictive Analytics, MLOps
Big Data & Processing	Databricks, Apache Spark, Apache Airflow, AWS Glue, AWS EMR, Delta Lake
Cloud Platforms	AWS (Bedrock, Glue, Textract, Transcribe, S3, Lambda, Redshift, EMR), GCP, Azure
Data Engineering	ETL/ELT Pipelines, Data Lakes, Data Warehousing, Data Modeling, Data Governance
Databases	Snowflake, Amazon Redshift, AWS RDS, MySQL, DynamoDB, MongoDB
DevOps & CI/CD	GitHub Actions, Jenkins, Terraform, Docker, Groovy Scripting
Orchestration	Apache Airflow, AWS Step Functions, AWS Lambda, EventBridge
Visualization	AWS QuickSight, SFMC Dashboards, Streamlit, Data-Driven Storytelling

CERTIFICATIONS

AWS Certified Data Engineer Associate	Databricks Certified Data Engineer Professional
Databricks Certified Generative AI Engineer Associate	Snowflake SnowPro Core Certified
HashiCorp Certified Terraform Associate (003)	GCP Professional Data Engineer
GCP Associate Data Practitioner	Azure Data Engineer Associate
Azure Fundamentals	Azure Data Fundamentals